Responsible Kubernetes:

clusters that don’t kill the planet (or your budget)



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About us

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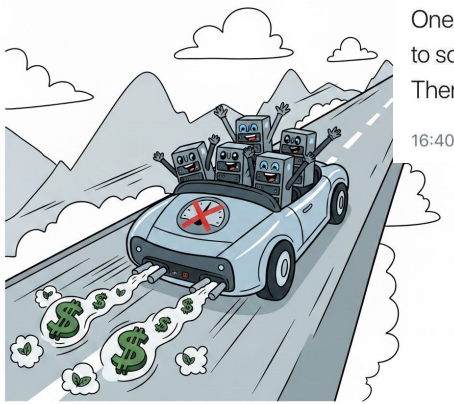
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At the beginning

- We didn’t think about costs or impact
- We were just writing code blindly
- We underestimate complexity and risks



SwiftOnSecurity
@SwiftOnSecurity

One time I tried to explain Kubernetes to someone.
Then we both didn't understand it.

16:40 · 06/08/2019 · Twitter for iPhone



A real problem ...



A real problem ...



A real problem ...



We thought we were heroes :/



Two different languages…

… and a new problem
that we have to face

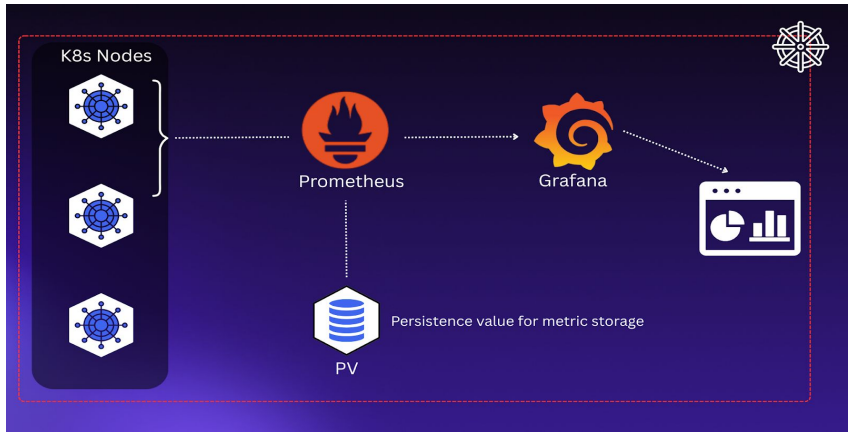
We can’t continue blindly !

- Why had the costs gone up so much?
- What exactly was causing the spike?
- What had we done?
- How could we fix it?

awslabs/**eks-node-viewer**

EKS Node Viewer

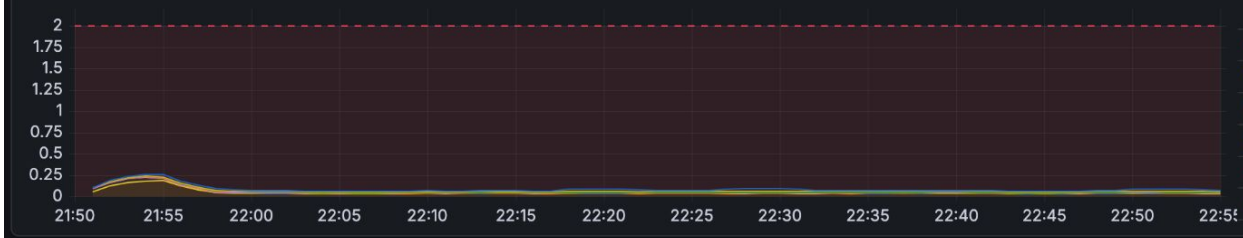
20 Contributors | 2 Used by | 2k Stars | 145 Forks





```
                                  memory                                  8%
ip-10-0-1-67.ec2.internal         cpu      5% (4 pods)  c5ad.2xlarge/$0.3440 On-Demand -        Ready         -
                                  memory   3%
                                  memory                                  6%
ip-10-0-5-23.ec2.internal         cpu      10% (4 pods)  c6a.xlarge/$0.1530  On-Demand - Ready -
                                  memory   6%
ip-10-0-6-158.ec2.internal        cpu      20% (5 pods)  i3.large/$0.1560    On-Demand - Ready -
                                  memory   3%
                                  memory                                  10%
ip-10-0-7-48.ec2.internal         cpu      20% (5 pods)  i3.large/$0.1560    On-Demand - Ready -
                                  memory   3%
```

CPU Usage

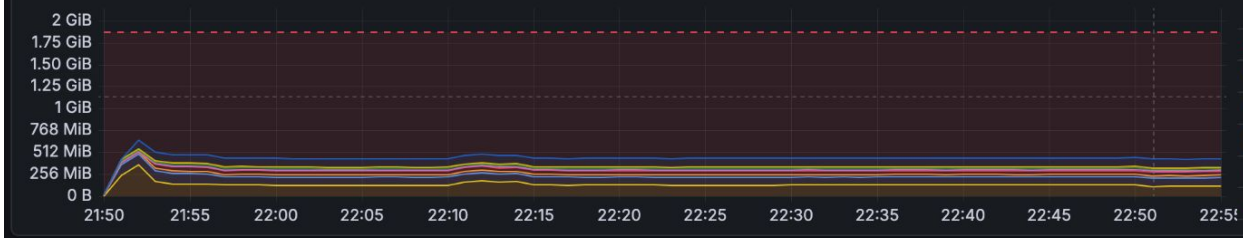


Name	Last *
max capacity	2
alloy-kw4nw	0.0343
aws-node-htdx9	0.00438
ebs-csi-node-r549v	0.00138
internal-ingress-controller-controller-6868db8645-p4p9q	0.00183
kube-prometheus-stack-prometheus-node-exporter-bjxvf	0.000566

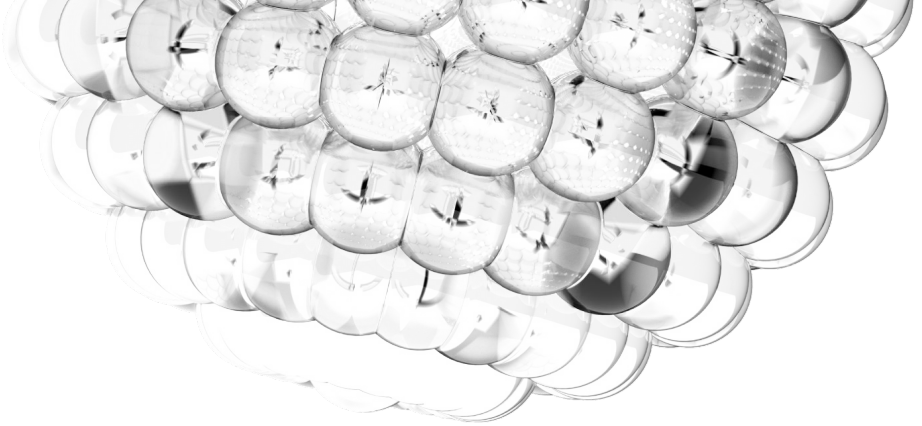
CPU Quota

Pod	CPU Usage	CPU Requests	CPU Requests %	CPU Limits	CPU Limits %
loki-canary-c9f87	0.0136				
alloy-kw4nw	0.0343	0.00100	3428%		
kube-proxy-mznqd	0.00161	0.100	1.61%		
kube-prometheus-stack-prometl	0.000566	0.200	0.283%	0.500	0.113%

Memory Usage (w/cache)



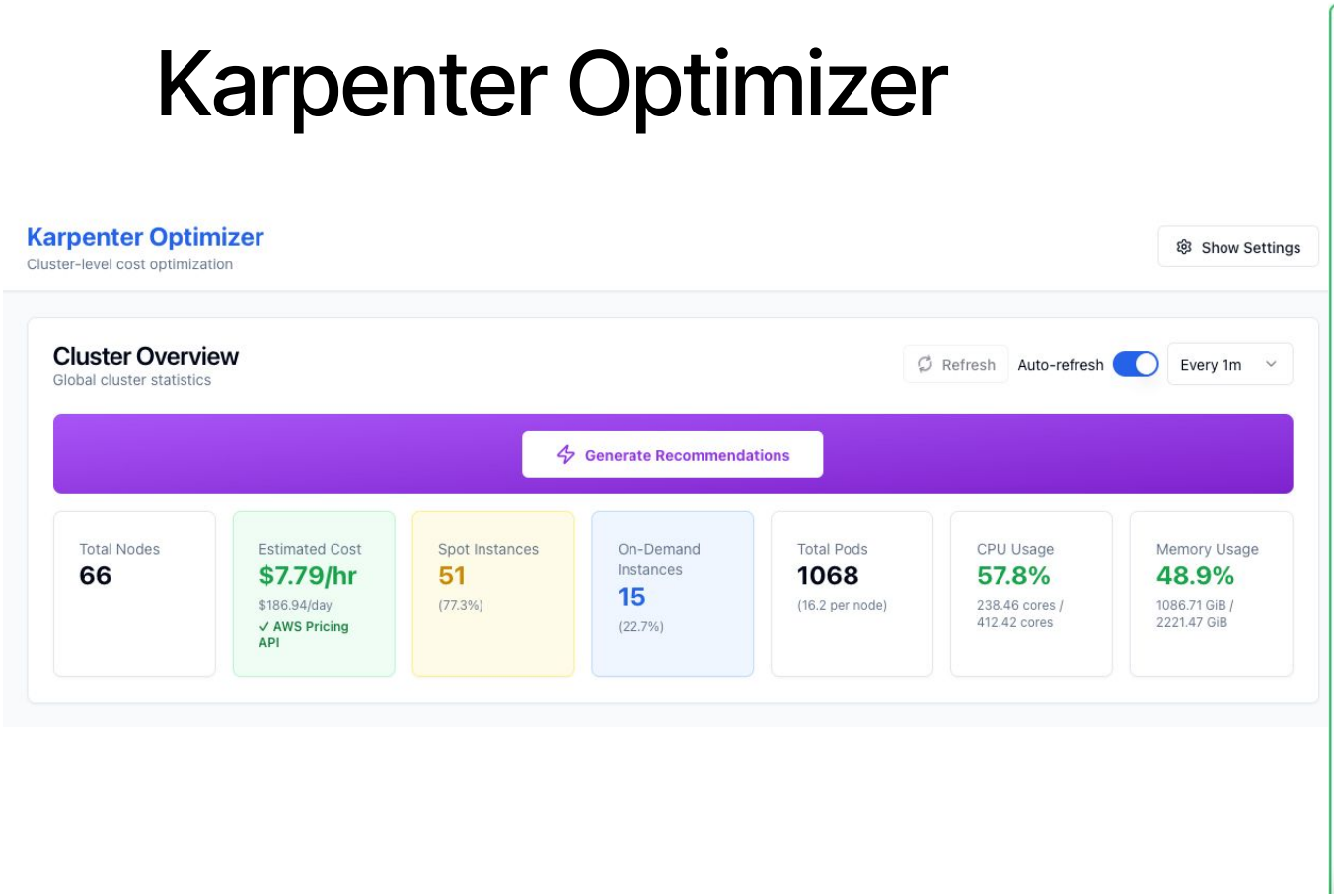
Name	Last *
max capacity	1.87 GiB
alloy-kw4nw	120 MiB
aws-node-htdx9	94.7 MiB
ebs-csi-node-r549v	30.1 MiB
internal-ingress-controller-controller-6868db8645-p4p9q	44.2 MiB
kube-prometheus-stack-prometheus-node-exporter-bjxvf	10.5 MiB

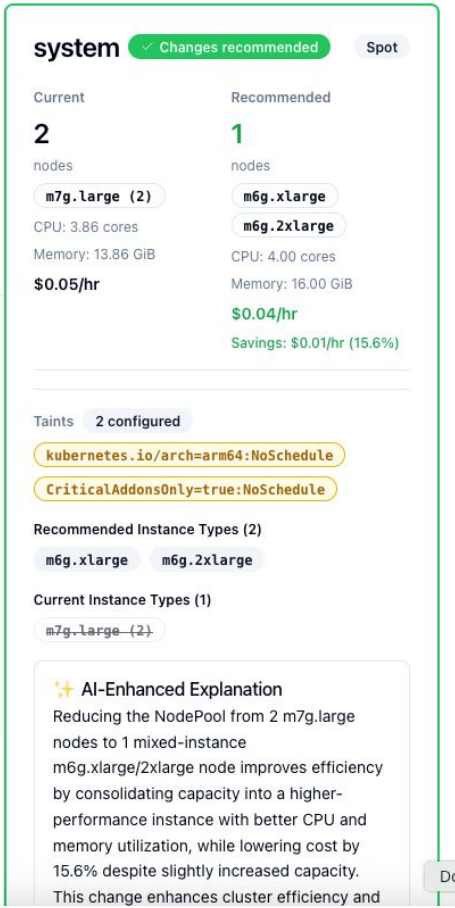


Other useful tool



Karpenter Optimizer





```yaml
apiVersion: karpenter.sh/v1
kind: NodePool
metadata:
  name: sftpgo
spec:
  template:
    metadata:
      labels:
        nodepool: sftpgo
        dedicated: sftpgo
    spec:
      requirements:
      - key: kubernetes.io/arch
        operator: In
        values: ["amd64"]

      - key: kubernetes.io/os
        operator: In
        values: ["linux"]

      - key: karpenter.sh/capacity-type
        operator: In
        values: ["on-demand"]

      - key: karpenter.k8s.aws/instance-category
        operator: In
        values: ["i", "d", "r"]

      - key: karpenter.k8s.aws/instance-generation
        operator: Gt
        values: ["4"]

      - key: karpenter.k8s.aws/instance-local-nvme
        operator: Gt
        values: ["500"]

      - key: karpenter.k8s.aws/instance-network-bandwidth
        operator: Gt
        values: ["10000"]

      nodeClassRef:
        group: karpenter.k8s.aws
        kind: EC2NodeClass
        name: default

      expireAfter: 1440h

      taints:
      - key: dedicated
        value: sftpgo
        effect: NoSchedule

  limits:
    cpu: 2000
    memory: 2000Gi

  disruption:
    consolidationPolicy: WhenEmpty
```



¿And how about Green0ps?

Well ... That's 🌟 " Green0ps " 🌟 too!



Why is now so important?

- Cloud makes infrastructure feel invisible, but the impact is still physical and real.
- Waste scales automatically if platform defaults are not optimized.
- Engineers now have direct impact on environmental footprint through daily decisions.



What would we have liked to know?



just because it works doesn’t mean it’s good.

we’re learning tools, not decisions.

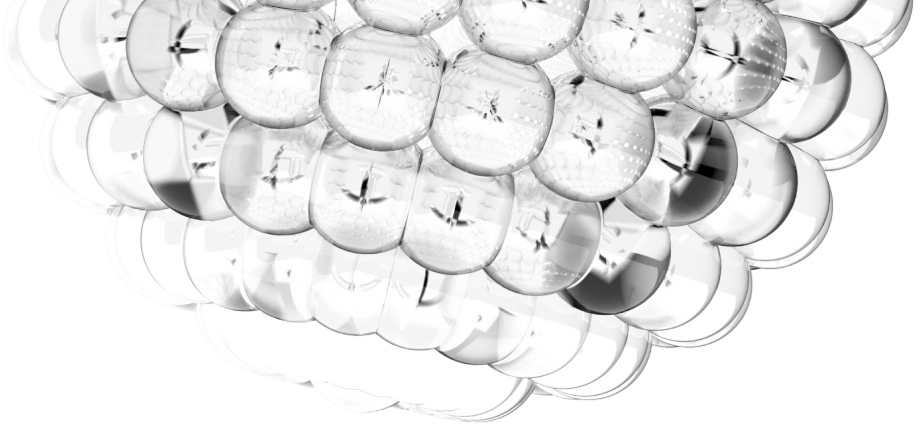
you cannot be sustainable alone.

your impact starts on day one.

What can you take away from this talk?



- If you're a junior: a couple of tips!
- If you're a senior: don't forget the basics!



Thanks!

Scan me!

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